



Republic of Rwanda
Ministry of Education



ACOUSTIC AND THERMAL INSULATIONS

BDCAT501

Perform Acoustic and Thermal Insulations

Competence

RQF Level: 5

Learning Hours



Credits: 6

Sector: Construction and Building Services

Trade: Building Construction

Module Type: specific

Curriculum: CBSBDC5001- TVET Certificate 5 in Building Construction

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Purpose statement	This specific module describes the knowledge, skills, and attitude of performing acoustic and thermal insulation works. It is intended to be pursued by the learners at TVET Certificate V in building construction. At the end of this module, the learner will be able to perform preliminary works, construct soundproof elements, and install thermal insulation panels.						
Learning assumed to be in place	Roof Construction, Ceiling Installation, Basic Reinforced Concrete Design, Scaffolding Operation, ARCHICAD software, Applied Physics						
Delivery modality	Training delivery			100%	Assessment		Total 100%
	Theoretical content			30%	Formative assessment	30%	50%
	Practical work:			70%		70%	
	• Group project and presentation	20%					
	• Individual project /Work	50%					
				Summative Assessment			50%

Elements of Competency and Performance Criteria

Elements of competency	Performance criteria	
1. Perform preliminary works	1.1.	Sound and thermal insulation requirements are identified based on the usage of the building
	1.2.	Materials, tools, and equipment are appropriately selected in accordance to the project acoustic insulation specifications.
	1.3.	Materials, tools, and equipment are appropriately selected in accordance to the project thermal insulation specifications.
	1.4.	Working place is properly prepared based on-site conditions and type of insulation
2. Construct sound proof elements	2.1.	Walls are systematically constructed according to the sound intensity required
	2.2.	Floors are systematically constructed according to the sound intensity required
	2.3.	Roof and ceiling are systematically constructed according to the sound intensity required

	2.4. Sound proof is accurately tested as per the required level of sound intensity
3. Install thermal insulation system	3.1. Thermal insulator frame components are properly assembled based on construction drawing details
	3.2. Thermal insulator materials are well applied based on their method of applications
	3.3. Thermal insulation system is accurately tested based on the required heat transfer

Intended Knowledge, Skills and Attitude

Knowledge	Skills	Attitude
✓ Describe sound and thermal insulation techniques ✓ Describe soundproof materials ✓ Describe thermal insulation materials ✓ Identify tools and equipment for thermal insulation ✓ Identify basic acoustic principles ✓ Describe regulations and standards for soundproofing system ✓ Explain acoustic and thermal safety measures	✓ Communicate at the workplace ✓ Interpret construction drawings ✓ Perform cutting techniques ✓ Install sound proof components ✓ Install thermal insulator components ✓ Perform assembling of thermal frame ✓ Perform fixing techniques of acoustic panels ✓ Perform fixing techniques of thermal insulators ✓ Perform sound test/acoustic test ✓ Perform finishing techniques of sound and thermal insulations	✓ Make the right decision within project execution ✓ Work under unexpected timeframes and conditions ✓ Pay attention to project details ✓ Work collaboratively with the team to achieve set project requirements ✓ Use resources effectively during soundproof execution ✓ Use creativity and innovation throughout the execution works ✓ Adapt to project changes during soundproofing works ✓ Meet set project deadlines ✓ Focus on project effectiveness

Course content

Learning outcomes	At the end of the module the learner will be able to: 1. Perform preliminary works 2. Construct sound proof elements 3. Install thermal insulation system
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Indicative content

- **Identification of sound and thermal insulation requirements**
 - ✓ Identify sound insulation requirements
 - + Sound propagation
 - + Noise Reduction Coefficient
 - + Impact Noise Insulation
 - + Sound standards
 - + Building Codes and Regulations
 - ✓ Analyze rate sound intensity
 - ✓ Identify behaviour of sound in an enclosed space
 - + Absorption
 - + Reflection
 - + Diffusion
 - ✓ Identify thermal Insulation requirements:
 - + Thermal transmittance
 - + Climate considerations
 - + Moisture management
 - + Building codes and regulations
 - ✓ Analyze temperature effect
- **Identification of acoustic materials, tools, and equipment**
 - ✓ Classification of acoustic materials
 - + Absorptive materials
 - + Diffusive materials
 - + Decorative active materials
 - + Hybrid materials
 - ✓ Classification of acoustic tools
 - ✓ Classification of acoustic equipment
 - ✓ Selection of acoustic materials
- **Identification of thermal materials, tools, and equipment**
 - ✓ Classification of thermal materials
 - ✓ Classification of thermal tools
 - ✓ Classification of thermal equipment
 - ✓ Selection of thermal materials
- **Preparation of Working place**
 - ✓ Clean the workplace
 - ✓ Protect the workplace
 - ✓ Erect scaffolding

Resources required for the indicative content

Equipment	Safety equipment (PPE), drilling machine, screw gun, staple gun, rivet gun, ladder, mitre saw, jig saw, nail gun, laser level, router machine, angle grinders, welding machine, Caulking Gun, Staple Gun, Ventilation Fan,
Materials	Nails, masking tape, sandpaper, timber beam, ply wood ,timber plank, sanding sealer, gypsum board, screws, nails, transparent sheets, acoustic ceiling panel, loose fill, rock wool, foams, fiberglass, resilient channels, acoustic hangers (mounts), soundproof drywall (plasterboard), acoustic sealants, acoustic putty, screws, J-Hook.
Tools	Sheet Metal Shears, Tape measure, laser meter, scrapper, spirit levels, hack saw, screwdriver kit, hand saw, chisels, universal pliers, pincer, scissor, utility knife, plumb bob, club hammer, claw hammer, T-square,
Facilitation techniques	Trainer guide, Simulation, individual work - The trainer provides a summary of the requirements for sound and thermal insulation in buildings and he /she asks the students to discuss these topics in groups and present their findings - The trainer provides a summary of the materials, tools, and equipment required for acoustic structures and he/she asks the students to participate in selecting these items based on their classification
Formative assessment methods	- Oral assessment - Written assessment - Performance assessment

Learning outcome 2: Construct soundproof elements

Learning hours: 25hours

Indicative content

- **Construction of sound proof walls**
 - ✓ Identification of wall construction materials
 - ✚ Based on physical properties
 - ✚ Based on materials composition
 - ✓ Identification of different types of acoustic walls
 - ✚ Rigid and massive homogeneous walls
 - ✚ Partitions of porous materials
 - ✚ Double wall partitions
 - ✚ Cavity wall Construction
 - ✚ Mineral Wool
 - ✓ Constructing acoustic walls
 - ✓ Finishing wall with sound proof element
- **Construction of sound proof floors**
 - ✓ Prepare Sub floor

- ✓ Underlay Soundproofing
- ✓ Install Flooring materials
- ✓ Isolate Perimeter
- **Construction of sound proof ceiling and roof**
 - ✓ Identification of roof techniques
 - ✓ Constructing acoustic Roof
 - + Prepare roof acoustic materials
 - + Evaluate roof structure
 - + Fix sound proofing layers
 - + Decoupling of sound
 - ✓ Identification of ceiling techniques
 - ✓ Constructing acoustic ceiling
 - + Prepare ceiling area
 - + Install ceiling framework
 - + Fix ceiling acoustic panels
 - + Apply finishing materials
- **Test Sound intensity**
 - ✓ Select test location
 - ✓ Identification of sound testing tools
 - ✓ Set up sound equipment
 - ✓ Create sound source
 - ✓ Record measurements
 - ✓ Analyse data
 - ✓ Produce report

Resources required for the indicative content

Equipment	Safety equipment (PPE), scaffolding, drilling machine, screw gun, staple gun, rivet gun, ladder, miter saw, jig saw, nail gun, laser level, router machine, angle grinders, welding machine, staples, Staple Gun, sound meter, Noise meter, drilling machine, concrete mixer, mortar mixer, concrete vibrator, wheelbarrow
Materials	Nails, Masking tape, sandpaper, timber beam, ply wood, timber plank, sanding sealer, gypsum board, screws, nails, transparent sheets, acoustic ceiling panel, loose fill, rock wool, foams, fiberglass, resilient channels, acoustic hangers (mounts), soundproof drywall (plasterboard), acoustic sealants, acoustic putty, cement, aggregate, paints, water, fuel, screwdriver, J-Hook,
Tools	Sheet Metal Shears, Tape measure, drawing set, laser meter, scrapper, spirit levels, hack saw, screw driver kit, hand saw, chisels, universal pliers, pincer, scissor, utility knife, plumb bob, club hammer, claw hammer, T-square, try-square, builder's square, marking gauge, building line, trowel, straight edge, hose pipe, decameter

Facilitation techniques	Trainer guide, Simulation, individual work, • The trainer displays a video demonstrating the construction of soundproof walls using various bonding techniques, then he/she requests the students to summarize the procedures involved in erecting such walls • Trainer and trainees perform the activity of constructing acoustic floors and he/she asks trainees to construct acoustic floors themselves • Trainer provides trainees site visits on ongoing project.
Formative assessment methods	• Oral assessment • Written assessment • Performance assessment

Learning outcome 3: Install thermal insulation system	Learning hours: 20 hours
Indicative content	
• Installation of Thermal insulator frame components ✓ Identify thermal insulator frame types ✓ Identify thermal insulation materials ✓ Identify thermal insulator frame components ✚ Header ✚ Studs ✚ Runners ✚ Sill plates ✚ Top plates ✚ Rafters ✓ Measuring frame components ✓ Cutting frame components ✓ Assemble and fix thermal insulation frame components • Apply Thermal insulation materials ✓ Check thermal insulation materials ✚ Mineral wools ✚ Fiberglass ✚ Foam board ✚ Prefabricated Concrete block ✓ Install thermal insulation materials ✚ Install Mineral wools ✚ Install Fiberglass ✚ Install Foam board ✚ Install Prefabricated Concrete block ✓ Place and fix thermal insulation materials	

- ✓ Seal thermal insulator material joints
- ✓ Apply finishes materials

- **Test Thermal insulation system**

- ✓ Identify thermal insulation testing types
 - ✚ Temperature cycling testing
 - ✚ Thermal shock testing
 - ✚ heat aging testing
 - ✚ Combined thermal and environmental testing
- ✓ Conduct thermal insulation testing
 - ✚ Temperature cycling testing
 - ✚ Thermal shock testing
 - ✚ heat aging testing
 - ✚ Combined thermal and environmental testing

Resources required for the indicative content

Equipment	Safety equipment (PPE), scaffolding, drilling machine, screw gun, staple gun, rivet gun, ladder, miter saw, jig saw, nail gun, laser level, router machine, angle grinders, welding machine staples, Staple Gun, Ventilation Fan, Caulking Gun, Thermometer
Materials	Nails, Masking tape, sandpaper, timber beam, ply wood, timber plank, sanding sealer, gypsum board, screws, nails, transparent sheets, acoustic ceiling panel, loose fill, rock wool, foams, fiberglass, resilient channels, acoustic hangers (mounts), soundproof drywall (plasterboard), acoustic sealants, acoustic putty, screws, J-Hook,
Tools	Sheet Metal Shears, Tape measure, drawing set, laser meter, scrapper, spirit levels, notebook, hack saw, screw driver kit, hand saw, chisels, universal pliers, pincer, scissor, utility knife, plumb bob, club hammer, claw hammer, T-square, try-square, builder's square, marking gauge
Facilitation techniques	Trainer guide, Simulation, individual work, - The trainer shows a video featuring a thermal insulator structure and he /she asks the students to discuss and summarise the steps required for installing thermal insulator materials. - The trainer prepares tasks for installing insulation panels to the structure and he /she asks to the students to perform the activity individually
Formative assessment methods	- Oral assessment - Written assessment - Performance assessment

Integrated/Summative assessment (For specific module)

Integrated situation

XG Hotel has a contract with RK Construction CO Ltd to construct an insulated Meeting room which has 4m x 3m x 3.20 m dimensions located in the north province, Musanze District in a single house according to the new implementation of sound & thermal insulation policies and regulations to avoid the noise and thermal caused by the loudspeakers, climate respectively in the meeting. As a technician from the foresaid Company, you are requested to:

- a) Identify the source of sound and high temperature.
- b) Construct cavity wall with 10 cm of thickness for void
- d) Install thermal insulation materials

The Meeting room is a new building shall be inaugurated after the insulation of this meeting room.

NB: all material and equipment shall be covered by XG Hotel.

Resources

Tools	Sheet Metal Shears, Tape measure, drawing set, laser meter, scrapper, spirit levels, hack saw, screwdriver, hand saw, chisels, universal pliers, pincer, scissor, utility knife, plumb bob, club hammer, claw hammer, T-square, try-square, builder's square, marking gauge
Equipment	Safety equipment (PPE), scaffolding, drilling machine, screw gun, staple gun, rivet gun, ladder, miter saw, jig saw, nail gun, laser level, router machine, angle grinders, welding machine, Staple Gun, Ventilation Fan, Caulking Gun,
Materials/ Consumables	Nails, Masking tape, sandpaper, timber beam, ply wood, timber plank, sanding sealer, gypsum board, screws, nails, transparent sheets, acoustic ceiling panel, loose fill, rock wool, foams, fiberglass, resilient channels, acoustic hangers (mounts), soundproof drywall (plasterboard), acoustic sealants, acoustic putty, screws, staples, J-Hook

Assessable outcomes	Assessment criteria (Based on performance criteria)	Indicator	Observation		Marks allocation
			Yes	No	
Learning outcome 1: Perform preliminary works	1.1. Sound and thermal insulation requirements are identified based on the usage of the building	Structure plan has been interpreted			3
		Source of sound is identified			3
		Source of heat is identified			3
	1.2. Materials, tools, and equipment are appropriately selected in accordance to the project acoustic insulation specifications.	Sound and thermal insulation Materials are selected			3
		Sound and thermal insulation tools are selected			3

		Sound and thermal insulation equipment are selected			3
	1.3. Working place is properly prepared based on-site conditions and type of insulation	Workplace is cleaned			3
		Workplace is protected			3
Learning outcome 2: Construct sound proof elements	2.1Walls are systematically constructed according to the sound intensity required	Cavity wall is constructed			6
		The type of bond is respected			4
		Acoustic materials are filled			5
		Finishes are applied			5
	2.4. Sound proof system is accurately tested as per the required level of sound intensity	Sound source is created			5
		Sound is recorded			5
		Sound is tested			5
		Sound is absorbed			5
Learning outcome 3: Install thermal insulation system	3.1. Thermal insulator frame components are properly assembled based on construction drawing details	Drawing is interpreted			6
		Insulator frame components are assembled			5
		Finishes are applied			5
	3.2. Thermal insulator materials are well applied based on their method of applications	Heat is tested			5
		Heat is absorbed			5
	3.3 Thermal insulation system is accurately tested based on the required heat transfer	Heat transfer is recorded			5
		Heat transfer is tested			5
	Total marks		100		
Percentage Weightage		100%			
Minimum Passing line % (Aggregate): 70%					

References

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